Linear Equations & Systems

ANSWER KEY

Solving single-variable linear equations

1 Solve for x: $3x - 7 = 20$.

$3x = 27$, so $x = 9$.

2 Solve for x: $\frac{2x + 5}{3} = 7$.

Multiply both sides by 3: $2x + 5 = 21$, so $2x = 16$, giving $x = 8$.

3 Solve for x: $5(x - 2) = 3(x + 4)$.

Expand: $5x - 10 = 3x + 12$. Subtract $3x$: $2x - 10 = 12$, so $2x = 22$, giving $x = 11$.

Equations with no solution or infinitely many solutions

4 For what value of k does the equation $4x + 6 = kx + 6$ have infinitely many solutions?

For infinitely many solutions, both sides must be identical for every x, so the coefficients of x must match: $k = 4$ (and the constants already match at $6 = 6$).

5 Explain why the equation $2x + 3 = 2x + 7$ has no solution, without solving it algebraically first.

Both sides have the same coefficient of x (2), so subtracting $2x$ from both sides leaves $3 = 7$, a false statement independent of x. Since no value of x can make a false numerical statement true, the equation has no solution.

Solving systems by substitution

6 Solve the system: $y = 2x + 1$ and $3x + y = 16$.

Substitute: $3x + (2x + 1) = 16$, so $5x + 1 = 16$, giving $5x = 15$, $x = 3$. Then $y = 2(3) + 1 = 7$. Solution: $(3, 7)$.

7 Solve the system: $x - 2y = 5$ and $2x + 3y = 3$.

From the first equation, $x = 5 + 2y$. Substitute into the second: $2(5 + 2y) + 3y = 3$, so $10 + 4y + 3y = 3$, giving $7y = -7$, $y = -1$. Then $x = 5 + 2(-1) = 3$. Solution: $(3, -1)$.

Solving systems by elimination

8 Solve the system: $2x + y = 11$ and $x - y = 1$, using elimination.

Adding the two equations directly eliminates y: $3x = 12$, so $x = 4$. Substituting into $x - y = 1$: $4 - y = 1$, so $y = 3$. Solution: $(4, 3)$.

9 Solve the system: $3x+2y=16$ and $5x-2y=8$, using elimination.

Adding the two equations eliminates y: $8x=24$, so $x=3$. Substituting into $3x+2y=16$: $9+2y=16$, so $2y=7$, giving $y=3.5$. Solution: $(3, 3.5)$.

Systems with no solution or infinitely many solutions

10 For what value of c does the system $6x+9y=12$ and $2x+3y=c$ have infinitely many solutions?

Dividing the first equation by 3 gives $2x+3y=4$. For the system to have infinitely many solutions, the second equation must represent the same line, so $c=4$.

11 Explain why the system $4x-2y=10$ and $2x-y=3$ has no solution.

Dividing the first equation by 2 gives $2x-y=5$. This directly contradicts the second equation, $2x-y=3$ — the same left-hand expression cannot equal two different values simultaneously, so the two lines are parallel (same slope, different intercepts) and never intersect.

Translating word problems into linear equations

12 A rectangle's length is 4 more than twice its width. If the perimeter is 68, find the width and length.

Let width $=w$, length $=2w+4$. Perimeter: $2(w)+2(2w+4)=68$, so $2w+4w+8=68$, giving $6w=60$, $w=10$. Length $=2(10)+4=24$.

13 A taxi charges a flat fee of 4 plus 2.50 per mile. If a ride costs 21.50, how many miles was the ride?

Let m be the number of miles: $4+2.5m=21.5$, so $2.5m=17.5$, giving $m=7$ miles.

Systems as word problems

14 A movie theater sells adult tickets for 12 and child tickets for 8. If 150 tickets were sold for a total of 1560, how many of each type were sold?

Let a = adult tickets, c = child tickets. $a+c=150$ and $12a+8c=1560$. From the first equation, $c=150-a$. Substitute: $12a+8(150-a)=1560$, so $12a+1200-8a=1560$, giving $4a=360$, $a=90$. Then $c=150-90=60$. So 90 adult tickets and 60 child tickets were sold.

15 Two numbers have a sum of 45 and a difference of 11. Find the two numbers.

Let the numbers be x and y with $x+y=45$ and $x-y=11$. Adding: $2x=56$, so $x=28$. Then $y=45-28=17$. The numbers are 28 and 17.

Interpreting the slope and intercept of a linear model

16 A gym membership costs $C = 25 + 15m$ dollars, where m is the number of months. Interpret the meaning of the numbers 25 and 15 in this context.

The 25 represents a fixed one-time joining fee (the cost when $m = 0$ months), and the 15 represents the monthly membership rate — each additional month increases the total cost by 15 dollars.

17 Two companies charge for printing: Company A charges $C = 0.10p + 20$ and Company B charges $C = 0.15p$, where p is the number of pages. Find the number of pages at which both companies charge the same amount, and state which company is cheaper for 500 pages.

Set equal: $0.10p + 20 = 0.15p$, so $20 = 0.05p$, giving $p = 400$ pages. For 500 pages: Company A charges $0.10(500) + 20 = 70$ dollars; Company B charges $0.15(500) = 75$ dollars. Company A is cheaper at 500 pages.

Equations with variables on both sides and fractions

18 Solve for x: $\frac{x}{3} + \frac{x}{4} = 7$.

Multiply every term by the least common denominator, 12: $4x + 3x = 84$, so $7x = 84$, giving $x = 12$.

19 Solve for x: $2(x + 3) - 5 = 3(x - 1) + 2$.

Expand: $2x + 6 - 5 = 3x - 3 + 2$, so $2x + 1 = 3x - 1$. Subtract $2x$: $1 = x - 1$, giving $x = 2$.

Systems with three unknowns represented as a word problem

20 A vending machine sells snacks for 1.50 dollars and drinks for 2 dollars. On one day it sold 45 items total for 79.50 dollars. Set up and solve a system to find the number of snacks and drinks sold.

Let s = snacks, d = drinks: $s + d = 45$ and $1.5s + 2d = 79.5$. From the first, $d = 45 - s$. Substitute: $1.5s + 2(45 - s) = 79.5$, so $1.5s + 90 - 2s = 79.5$, giving $-0.5s = -10.5$, so $s = 21$. Then $d = 45 - 21 = 24$. So 21 snacks and 24 drinks were sold.

Choosing the best method for a given system

21 For the system $y = 4x - 3$ and $5x + 2y = 17$, explain which method (substitution or elimination) is more efficient here, and solve the system.

Substitution is more efficient here, since y is already isolated in the first equation. Substitute into the second: $5x + 2(4x - 3) = 17$, so $5x + 8x - 6 = 17$, giving $13x = 23$, $x = \frac{23}{13}$. Then $y = 4\left(\frac{23}{13}\right) - 3 = \frac{92}{13} - \frac{39}{13} = \frac{53}{13}$.

Consistent, inconsistent, and dependent systems

22 Classify the system $y = 3x + 2$ and $y = 3x - 4$ as consistent, inconsistent, or dependent, and explain your reasoning without solving.

Both lines have the same slope (3) but different y-intercepts (2 and -4), so they are parallel and never intersect. The system is inconsistent (no solution).